Exercise 2:

Let us consider the SuperStore Sales dataset found on Kaggle.

1. Create a notebook and load the dataset.
2. Display general information about the data set: (a) columns, (b) number of rows, (c) averaged numbers, (d) max/min values, and (c) the quality of data (number of missing values per column).
3. Additionally, Display a portion of the data (say the 10 first rows).
4. Disply the number of (a) unique values per column, (b) the type of values, and to group the dataset by a specific column say (column = order date), where (c) other values should be averaged, than (d) we extract the row having the highest average.
5. Rearrange the data so important columns are displayed in first.
6. To avoid all kind of type issues, we like to (a) convert all numerical values to float (prices, quantities …etc), and (b) all string values should be lowercased.
7. Codify sales into new column named "sale interest" which consists of three values (1: basic, 2: moderate, 3: high).
8. We like to normalize sale by dividing on the highest sale value (keep values in 0 to 1).
9. As to conserve privacy, we would to delete postal code and customer name from the data before passing it to other members in the team.
10. Delete duplicated records.
11. How many days we have in the studied period, how many weeks?
12. Check how many chip modes we have.
13. Check the customer who requests the most, the customer who spends the most, and the one who cares more about his deliveries, what is the portion of these users among the total users.
14. Calculate the period between order date and ship date.
15. Group products by category, and cite which category has the highest sales? Is the number of concerned customers considerable?
16. To help the marketing department, analyze the Segment column and suggest which segment should have more focus.
17. We like to improve the canning process, look at the request dates, and products to conduct if there are any product that needs a better canning.
18. We filter out all records in the period [X,Y] and calculate the sales average.
19. We filter out all records having (column=sale has value less than Y), then rows group by city and tell which city should have lower priorities.
20. For the city= X, we count how many times column= sale has value more than Y.
21. We interest by analyzing the products quantities, and their related sales, analyze both columns and assist the stock manager to avoid the lack of product (find out products average requests per day).
22. We interest by studying the change in the sales (days, weeks), create and display a series representing the difference in sales between each successive (a) two days (grouped by region), (b) two weeks.
23. The customer offered a new amount of data –of the same structure- for study and we like to insert it, (a) how to do that? (b) Ensure that indexes are ordered.